Base Research Paper & Original System

Base paper: “FPGA-Based Wild Animal Detection System Using Verilog: Advancing Surveillance and Mitigating Human-Wildlife Conflicts”
K. V. Abhishek and Rajesh Kannan Megalingam, ICCPCT 2025

1. Abstract of the Base Paper

The base paper presents an FPGA-based wild-animal detection system intended for real-time wildlife monitoring and human-wildlife conflict mitigation. The system uses a modular hardware architecture implemented in Verilog and combines Sobel edge detection, frame differencing for motion analysis, and predefined size and motion thresholds for object classification. A parallel-processing approach is used to improve computational performance. The paper validates the design through simulation in ModelSim using low-resolution grayscale video frames prepared with OpenCV. The reported system is designed to detect moving animal-like objects while aiming for low latency, low power and suitability for resource-constrained surveillance applications.

2. Main Ideas Already Present in the Base Paper

Base idea	What the paper does
Grayscale image input	Surveillance video is converted/preprocessed into low-resolution grayscale frames for hardware processing.
Sobel edge detection	A Sobel filter highlights object boundaries and intensity changes.
Frame differencing	Current and previous frames are compared to identify significant pixel changes caused by motion.
Motion / background analysis	Dynamic regions are isolated while static regions are reduced or filtered.
Threshold-based classification	Predefined size, shape and motion-related thresholds are used to distinguish relevant animal-like objects from irrelevant regions.
Parallel FPGA processing	Verilog modules operate concurrently under a common clock to exploit FPGA hardware parallelism.
Alert / control logic	Processing outputs are combined and an alert can be triggered when detection conditions are satisfied.

3. Base Processing Flow

Video / image frames → **Grayscale** → **Sobel edge detection + Frame differencing** → **Motion/edge information** → **Threshold-based object classification** → **Control / alert**

Important: The four items on the next page are proposed additions to the base system for our project. They are separated from the original paper’s features so the project clearly shows what is reproduced and what is added.

Proposed Novelty Ideas Added to the Base System

Four practical additions selected for implementation in Xilinx Vivado / Verilog

1. Parallel / Improved Sobel

Purpose: The base system already uses Sobel edge detection. Our addition is to make the Sobel stage more hardware-parallel and efficient.

Vivado / Verilog implementation: Use a 3×3 line-buffer/window structure and calculate horizontal (Gx) and vertical (Gy) gradients in parallel. Pipeline the arithmetic so different pixels are processed on successive clock cycles. An optional eight-directional Sobel version can be explored, but parallel Gx/Gy is the simpler implementation.

Expected benefit: Higher throughput and a clear hardware-parallelism contribution.

2. Shadow / False-Detection Suppression

Purpose: Frame differencing can react to shadows, illumination changes or short-lived motion. A simple temporal decision stage can reduce unstable detections.

Vivado / Verilog implementation: Add a small persistence counter/history register after motion thresholding. Require a region to remain detected for a selected number of consecutive frames before accepting it as valid. Combine this with minimum motion-pixel or object-size thresholds.

Expected benefit: Fewer short-lived or unstable false detections using simple counters, registers and comparators.

3. Bounding Box + Centroid Detection

Purpose: The base system identifies moving/object-like regions; our project can additionally localize the detected object and show where it is in the frame.

Vivado / Verilog implementation: For foreground pixels, maintain Xmin, Xmax, Ymin and Ymax registers. Also accumulate pixel count and X/Y coordinate sums. The centroid is obtained from the coordinate sums divided by the detected-pixel count. The bounding box can be displayed or used by later logic.

Expected benefit: Direct object localization and a much clearer FPGA demonstration.

4. Lightweight Animal-vs-Non-Animal Classifier

Purpose: The base paper uses predefined thresholds; our project can make the final decision more explicit by combining several simple geometric and motion features.

Vivado / Verilog implementation: Use bounding-box width, height, area/pixel count, aspect ratio, motion persistence and optionally fill ratio. Compare them with experimentally selected thresholds using comparators and simple rule-based logic. Output: Animal-like or Other / Non-animal.

Expected benefit: Adds practical classification without requiring a CNN, GPU or large machine-learning model.

Proposed Final Project Architecture

Input Frame → **Grayscale** → [**Frame Difference** → Motion Threshold] + [**Parallel Sobel**] → **Fusion** → **Shadow / Noise Suppression** → **Object Region** → **Bounding Box + Centroid** → **Lightweight Classifier** → **Animal-like / Other**

Implementation strategy: First reproduce the base-paper pipeline in Vivado. Then add the four novelty blocks one at a time and compare outputs before and after each addition. This clearly separates the base work from our contribution.

Source basis: Abhishek & Megalingam, ICCPCT 2025. The parallel/improved Sobel suggestion is additionally informed by Ravichandran et al., Sensors 2025.